MIL-L-22692A (OS)
7 August 1968
Superseding
MIL-L-22692 (Wep)
24 October 1960

MILITARY SPECIFICATION

FOR

LOCATOR, ORDNANCE, MARK 9 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification describes the requirements for the procurement of the Locator, Ordnance, Mark 9 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, or as noted, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-16775	Chests, Field, Aluminum, Empty

Federal

PPP-B-636	Boxes, Fiber
QQ-S-781	Strapping, Flat; Steel

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	FUZE AND FUZE COMPONENTS, ENVIRONMENTAL AND PERFORMANCE TESTS FOR

DRAWINGS

Navy Department

FSC 6665

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BUORD LD

No. 480240	Locator, Ordnance, Mark 9 Mod 0
No. 496307	Transit Case Assembly

BUORD Dwg.

No. 1827544	Target Test Block
No. 1827545	Phase Test Block
No. 1827516	Connector
No. 1827542	Plug Extra Gland
No. 1875134	Assembly
No. 1698359	Search Head Assembly
No. 2007343	Special Battery Compartment Cover
No. 2293749	Head Set

PUBLICATIONS

Naval Ordnance Systems Command

OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
OD 10511	Basic Methods for Nonmagnetic Testing with the Test Set Mark 115 Type

(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and Supplements	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20360.)

3. REQUIREMENTS

3.1 Description - The device specified herein is a portable locator for locating metallic objects. It is designed to withstand submergence and to operate in water to a depth of 100 feet. The locator consists of five components: the control unit, the search head, search head handle, web belt, and the headset. The transit case which is provided holds the locator, a set of five spare plug-in units, two test blocks, and a double male connector. It also has provisions for temporary storage of one set of batteries. The locator is carried by a swimmer with the control unit attached to a web belt worn around the waist and may be used in conjunction with most types of breathing apparatus in use by the fleet. The swimmer searches for ordnance with the search head while listening to the sound in the headset to indicate detection.

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3.2 General Requirements

3.2.1 Preproduction Samples - Unless otherwise specified in the contract or order, preproduction samples of the locator shall be manufactured using the specific methods, tools, and procedures intended for the production lot. These samples will be tested as specified in Section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision for samples shall be in accordance with 4.1.1

3.2.2 Test Procedures - The procedures of 4.5.4 shall be carried out completely for any tests.

3.2.3 The target test block shall meet the requirements of BuOrd Dwg. 1827544.

3.2.4 The phase test block shall meet the requirements of BuOrd Dwg. 1827545.

3.2.5 Materials - Unless otherwise specified in the contract, requisition, or order; all materials and purchased parts entering into the final assembly of the locator shall conform to the detailed requirements indicated on the drawings and specifications listed in BuOrd List of Drawings LD No. 480240.

3.2.6 Finishes, Coating, Construction, and Dimensions - Finishes, coatings, construction, and dimensions shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BuOrd List of Drawings LD No. 480240.

3.2.7 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BuOrd List of Drawings LD No. 480240.

3.2.8 Interchangeability - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings, shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required then changes in dimensions are necessary, and revision and correction of the applicable drawings is required. (See 3.2.12.)

3.2.9 Gages - The contractor shall provide whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition; the government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.10 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the GO and NOT GO gaging practice shall be as set forth in Handbook H28.

3.2.11 Requests for Deviation - Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be as set forth in OP 400.

3.2.12 Conflicting Requirements - Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

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3.3 Performance Requirements and Product Characteristics - The locator shall meet the following performance requirements and product characteristics:

3.3.1 Preconditioning - All locators (less batteries) shall be preconditioned during manufacture according to the procedures of 4.5.4.2.

3.3.2 Submergence and Leakage Resistance - The locator shall be subjected in the following order to the preconditioning cycles of 4.5.4.2, the sea water submergence test specified in 4.5.4.3, and to the search head and cable leakage resistance test of 4.5.4.4, after which it shall meet all the performance and other requirements of this specification.

3.3.3 Oscillator Frequency Adjustment Range - The frequency of the oscillator circuit shall be adjustable from at least 10 cps below 2500 cps to at least 10 cps above 2500 cps when tested as specified in 4.5.4.5. The oscillator frequency of all locators shall be factory adjusted to 2500 ± 3 cps.

3.3.4 Front Panel Controls - When the locator is tested as specified in 4.5.4.7, the R front-panel control shall be less than ±360° from the center of its 5 turn rotational travel and the X front panel control shall be less than ±45° from the center of its rotational travel after both controls have been adjusted for the minimum (i.e. null voltage) across the headset terminals.

3.3.5 Operational Requirements - The locator, when tested as specified in 4.5.4.8 and 4.6.10 shall meet the following operational requirements:

3.3.5.1 The rms voltage, measured across the headset terminals, shall be within the ranges specified in table 1 for the corresponding combinations of ambient temperature and battery voltage conditions as shown in the table, when the standardized target pin (4.4.2) is positioned over either of the receive coils of the search head and with the "OFF-HI-LO" switch set to HI.

3.3.5.2 With the "OFF-HI-LO" switch set to HI, the minimum rms voltage (i.e., null voltage) measured across the headset terminals of each locator subjected to the test as shown in table 1 shall be 40 millivolts maximum.

3.3.5.3 The rms voltages measured across the headset terminals, for each test condition "A", "E", "H", "K", and "N" of table 1, with the standardized target pin (4.4.2) positioned over the left receive coil and with the "OFF-HI-LO" switch on HI, shall be within ±1db of the voltage measured under the same conditions but with the standardized target pin (4.4.2) positioned over the right receive coil.

TABLE 1

Test Condition	Ambient Temperature	"A" Voltage	"B" Voltage	RMS Millivolt Level Across Headset Terminals
A	120°F ± 5°F	1.25 ± 0.05	56 ± 0.5	450 - 1000
B	120°F ± 5°F	1.40 ± 0.05	63 ± 0.5	450 - 1400
C	120°F ± 5°F	1.10 ± 0.05	48 ± 0.5	250 - 1000
D	95°F ± 5°F	1.40 ± 0.05	63 ± 0.5	450 - 1300
E	95°F ± 5°F	1.25 ± 0.05	56 ± 0.5	450 - 1000
F	95°F ± 5°F	1.10 ± 0.05	48 ± 0.5	250 - 1000
G	75°F ± 5°F	1.40 ± 0.05	63 ± 0.5	450 - 1200
H	75°F ± 5°F	1.25 ± 0.05	56 ± 0.5	450 - 650
I	75°F ± 5°F	1.10 ± 0.05	48 ± 0.5	250 - 650
J	55°F ± 5°F	1.40 ± 0.05	63 ± 0.5	450 - 1200
K	55°F ± 5°F	1.25 ± 0.05	56 ± 0.5	450 - 1000
L	55°F ± 5°F	1.10 ± 0.05	48 ± 0.5	225 - 1000
M	35°F ± 5°F	1.40 ± 0.05	63 ± 0.5	400 - 1200
N	35°F ± 5°F	1.25 ± 0.05	56 ± 0.5	400 - 1000
O	35°F ± 5°F	1.10 ± 0.05	48 ± 0.5	200 - 1000

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3.3.5.4 For each test condition "A", "E", "K:, and "N" of table 1, and with the standardized target pin (4.4.2) positioned over either receive coil, the rms voltage measured across the headset terminals with the "OFF-HI-LO" switch on LO, shall be between 25 and 85 millivolts. For condition "H" the voltage measured shall be between 30 to 80 millivolts.

3.3.5.5 The rms voltage measured across the headset terminals, with the phase test block (4.4.3) positioned over either of the receive coils and with the "OFF-HI-LO" switch on HI, shall be 7 db or more below the rms voltage measured when the standardized target pin (4.4.2) was placed over the same receive coil. This requirement shall hold for each combination of ambient temperature and battery voltage settings specified in table 1.

3.3.5.6 For each test condition "A", "E", "H", "K", and "N" of table 1, the rms voltage measured across the headset terminals with the phase test block (4.4.3) positioned over the left receive coil shall be within 2 db of the rms voltage measured under the same conditions except with the phase test block (4.4.3) positioned over the right receive coil.

3.3.5.7 For each test condition "A", "E", "H", "K", and "N", and with the "OFF-HI-LO" switch set to HI, any saturating target that is slowly brought close to either end of the search head, shall cause the headset voltage to increase by 10 db or more above the headset voltage that was measured when the standardized target pin (4.4.2) was positioned over either of the receive coils under the same test condition. As the saturating target is brought toward the search head, the headset voltage shall increase to a maximum at the search head or just prior to reaching the search head. If the maximum voltage occurs at some position of the target other than at the search head, the voltage with the target positioned on the search head shall not be more than 0.5 db below the maximum observed.

3.3.5.8 For each test condition "A", "E", "H", "K", and "N" of table 1, the null voltage measured across the headset terminals shall not exceed 80 millivolts when tested in accordance with 4.5.4.8.1.8.

3.3.6 Control Unit Temperature Stability

3.3.6.1 With the temperature of the search head maintained at 75 ± 5°F and with "A" and "B" battery voltages maintained at 1.25 ± 0.05 volts and 56 ± 0.5 volts respectively, the null voltage measured at the headset terminals (in accordance with 4.6.9) shall not vary more than ±8 db during the first 5-minute period after the ambient temperature of the control unit has been subjected to each of the following step changes of temperature -

From 120°F to 110°F ± 2°F

From 70°F to 60°F ± 2°F

From 42°F to 32°F ± 2°F

3.3.7 Electronic Circuit Requirements - When tested as specified in 4.5.4.9 and 4.6.11, the electronic circuits of the locator shall meet the requirements of 3.3.7.1 through 3.3.7.3.6 with "A" and "B" battery voltages maintained at 1.25 ± 0.05 volts and 56 ± 0.5 volts respectively, in an ambient temperature of 75 ± 5°F, and at a maximum relative humidity of 70 percent.

3.3.7.1 Frequency Response Output Amplifier and Transformer - The frequency response at 5000 cps shall be 60 db or more below the response at 2500 cps when tested in accordance with 4.6.11.1.

3.3.7.2 Second Harmonic Level at Test Point E-1 - The second harmonic voltage at test point E-1 when tested in accordance with 4.5.4.9, shall not exceed 200 millivolts rms.

3.3.7.3 Crosstalk Requirements - The locator shall meet the following crosstalk requirements when tested in accordance with 4.6.11.2.

3.3.7.3.1 The voltage measured across the headset terminals shall not exceed 10 millivolts when test point E-1 is shorted to ground.

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3.3.7.3.2 The voltage measured across the headset terminals shall not exceed 500 millivolts when the input grid lead to Z-6 is shorted to ground.

3.3.7.3.3 The voltage measured across the headset terminals shall not exceed 500 millivolts when the input grid lead to Z-6 is shorted to ground and Z-7 is removed from its socket.

3.3.7.3.4 The voltage measured across the headset terminals shall not exceed 300 millivolts when both Z-6 and Z-7 are removed from their sockets.

3.3.7.3.5 The voltage measured across the headset terminals shall not exceed 300 millivolts when only Z-6 is removed from its socket.

3.3.7.3.6 The voltage measured across the headset terminals shall not exceed 1 volt when the input lead to Z-5 is shorted to ground, and all other units are restored to the circuit.

3.3.8 Nonmagnetic Effects - The locator and all of its parts shall be nonmagnetic and shall be tested in accordance with the tests of 4.5.4.10.

3.3.8.1 The magnetic influence on each headphone shall not exceed 0.4 milligauss when tested as specified in 4.5.4.10.1.2 and the magnetic influence on each headband shall not exceed 0.05 milligauss when tested as specified in 4.5.4.10.1.1.

3.3.8.2 The magnetic influence on the search head shall not exceed 0.10 milligauss when tested as specified in 4.5.4.10.2.

3.3.8.3 The magnetic influence of the short handle coupled to the extension handle (collapsed) shall not exceed 0.10 milligauss when tested as specified in 4.5.4.10.3.

3.3.8.4 The magnetic influence of the control unit shall not exceed 4.0 milligauss when the control unit is idealized and 1.0 milligauss when the control unit is demagnetized, when tested as specified in 4.5.4.10.4.

3.3.8.5 The magnetic influence of the belt shall not exceed 0.10 milligauss when tested as specified in 4.5.4.10.5.

3.3.9 Spare Plug-In Units - The locator with its spare plug-in units installed shall meet all the requirements of this specification and all pertinent drawings.

3.3.10 Search Head - The search head shall meet all the requirements of drawing No. 1698359 and the magnetic requirements of 3.3.8.

3.3.11 Headset - The headset shall meet all the requirements of drawing No. 2293749 and the nonmagnetic requirements of 3.3.8.

3.4 Environmental Requirements -

3.4.1 Vibration; Transportation, Storage and Handling - The locator, in its transit case, shall withstand the vibration tests specified in 4.6.1, after which it shall show no damage and shall meet all the performance and other requirements of this specification.

3.4.2 Repeated Impact; Transportation, Storage and Handling - The locator shall withstand the test specified in 4.6.2 after which the locator shall show no damage and shall meet all the performance and other requirements of this specification.

3.4.3 Shock; Transportation, Storage and Handling - The locator shall withstand the shock tests specified in 4.6.3, after which the locator shall show no damage and shall meet all the performance and other requirements of this specification.

3.4.4 Temperature and Humidity - The locator (less batteries) shall be subjected to the temperature and humidity test of 4.6.4 after which it shall meet all the performance and other requirements of this specification.

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3.4.5 Submergence and Leakage Resistance - The locator (less batteries) shall be subjected to the submergence and leakage resistance tests of 4.6.5 after which it shall meet all the performance and other requirements of this specification.

3.4.6 Altitude - The locator, in its transit case, shall be subjected to the simulated altitude test corresponding to 50,000 feet altitude specified in 4.6.6 for a period of 15 minutes or more after which the locator shall meet all the performance and other requirements of this specification.

3.4.7 Salt Spray - The locator, removed from its transit case, and the empty transit case shall be subjected to the salt spray test specified in 4.6.7 after which the locator and its transit case shall show no damage or corrosion and shall meet all the performance and other requirements of this specification.

3.4.8 Cable Flexibility, Insulation, Low Temperature - Samples of the cables shall be subjected to the coil bend test specified in 4.6.8 after which they shall be given sea water pressure tests and cable leakage resistance tests. There shall be no evidence of damage to sheath, insulation, fillers or conductors.

3.4.9 Storage Life - The locator, stored in its transit case (but with the batteries removed), shall be operable and meet the performance and other requirements of this specification after being stored under normal conditions of storage for a period of at least 5 years, during which extremes of temperature from -20 to 160°F and humidity up to 95 percent relative, maximum, may be encountered. This requirement shall be considered as having been met when the locator has satisfied the requirements of 3.3 through 3.4.8.

3.5 Clean-Up - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spatter or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship - The equipment, including parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage and freedom of parts from burrs and sharp edges. Conductors shall not be damaged as a result of stripping of insulation or crimping of terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preproduction Sample - Unless otherwise specified for each contract or order, a representative sample of four (4) locators shall be manufactured using specific methods, tools, and procedures intended for the production lot. Also, a 10-ft. sample of search head cable and a 10-ft. sample of headset cable shall be selected for destructive tests. One special locator battery compartment cover (4.4.7) shall accompany the preproduction samples. These items shall be given the contractor's inspection (4.2) and the approved quality conformance tests defined in 4.3.1 and specified in 4.5. These items, plus a full report of the quality conformance test results and procedures, shall be forwarded by the Government Inspector to the activity designated in the contract, or by the Naval Ordnance Systems Command, for the purpose of determining compliance of the preproduction sample with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples, except search head cable samples, that are accepted will be applied as part of the quantity specified by the contract or order.

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4.1.2 Lot Size - As applied to inspection of units of product the term "lot" shall mean "inspection lot", i.e. a collection of units of product submitted by a supplier for testing in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.1.3 Periodic Production Samples - When specifically required by the Naval Ordnance Systems Command or its designated testing activity, a sample of locators will be selected at random as specified in 4.1.2 plus a 10-foot sample of each production lot of search head cable and headset cable, for submission to the Naval Ordnance Systems Command designated activity to determine compliance of the samples with all the requirements of the contract, specifications and drawings. The locator shall be considered satisfactory if it meets all the pertinent drawing and contract requirements, and the specification requirements of 3 through 3.6, inclusive.

4.2 Contractor's Inspection - Unless otherwise specified, the contractor shall perform all tests and inspections listed in this specification, and as otherwise required, to determine that the locator conforms to all applicable requirements.

4.3 Classification of Tests - The inspection and testing of the locator shall be classified as follows:

4.3.1 Quality Conformance Tests - These tests are to be accomplished on each locator being submitted for acceptance under contract. Inspection and test procedures shall be submitted for government approval prior to starting production. Quality conformance tests shall be performed by the manufacturer. These tests are specified in 4.5.

4.3.2 Preproduction and Periodic Production Tests - Preproduction and periodic tests are those which are performed on samples selected as specified in 4.1.1 and 4.1.3 which are representative of the production of the locator after the award of the contract to determine that the lot and production meet all the requirements of this specification. These tests are detailed in 4.6 and will be performed by an agency designated by he procuring activity. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with all requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test Site and Equipment - The following test site and equipment are required to perform the quality conformance tests set forth in this specification:

4.4.1 Suitable test site which meets the test requirements of 4.5.4.1.

4.4.2 Standardized target pin, see figure 1.

4.4.3 Phase test block, see drawing No. 1827545.

4.4.4 A completely nonmetallic chamber or cabinet, an open nonmetallic water tank, or a large room, in which the temperature can be controlled to within ±2°F over the range of 32 to 125° Fahrenheit. Any chamber, cabinet or water tank selected must be sufficiently large to contain an operating locator, and must satisfy the test site requirements of 4.5.4.1.

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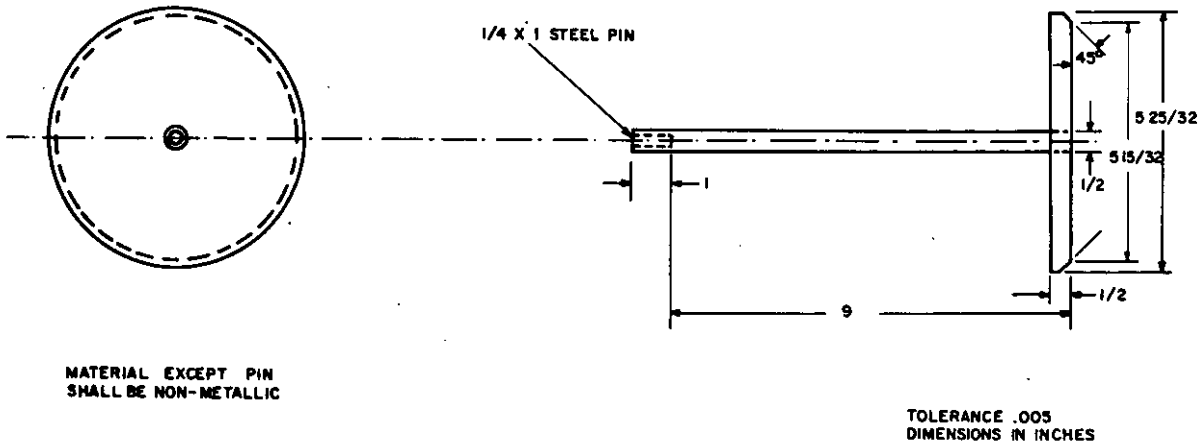


Figure 1. Standardized Target Pin

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4.4.5 A chamber or cabinet in which the temperature can be controlled to within ±5°F over the temperature range of 160 to -20°F and in which the temperature can be reduced linearly within 1 hour from 160 to-20° Fahrenheit. The chamber or cabinet must be sufficiently large to contain a nonoperating locator.

4.4.6 Suitable water tank sufficiently large to contain the control unit (two required).

4.4.7 Special locator battery compartment cover, drawing No. 2007343, with a watertight gland passing the battery cable into the control unit.

4.4.8 An a-c vacuum tube voltmeter having an input impedance of 10 megohms or more shunted by no more than 8 mmfs including any probe. The meter shall be operable over a frequency range of 1,000 to 10,000 cps, and shall be capable of measuring voltages from 0.001 to 30 volts (full scale) to an accuracy of 3 percent or better.

4.4.9 An audio frequency microvolter 1,000 to 10,000 cps, with an output voltage range of 0.5 microvolts to 1.0 volt with an accuracy of 3 percent or better.

4.4.10 An audio frequency signal generator capable of producing output voltages from less than 1 volt to 30 volts or more across a load impedance of 600 ohms over a frequency range of 1,000 to 10,000 cps.

4.4.11 Electronic counter (events-per-unit-time or EPUT meter) capable of counting events from 1,000 to 10,000 per second to an accuracy of ±1 event per second, and having a voltage sensitivity greater than 0.2 volts rms.

4.4.12 Voltmeter, a-c, 1000 ohms per volt or better, capable of measuring voltages from 0.01 to 3 volts (full scale) with an accuracy of 2 percent or greater over a frequency range of 100 to 15,000 cps.

4.4.13 D-C power supplies shall be used to furnish the "A" and "B" power requirements. The two adjustable "A" power supplies shall be used to individually furnish power to the two "A" filament circuits. Each of the three power supplies shall be adjustable over the specified range and shall regulate for 10 percent line variations and load variations within 0.1 percent when supplying up to 100 ma for the "A" power supplies and up to 20 ma for the "B" power supply. The ripple output shall be less than 0.2 millivolts rms. The power supplies are connected to the locator with cable drawing No. 2007343. The battery circuit grounds are connected to the locator ground only through the on-off switches.

4.4.14 Suitable water pressure tank capable of containing the assembled locator, and simulating sea water depths equivalent to 100 feet.

4.4.15 Megohm bridge capable of measuring leakage resistance up to 1,000,000 megohms at a potential of 100 volts with a accuracy of ±30 percent or better.

4.5 Quality Conformance Tests

4.5.1 Visual and Dimensional Examination - Each item shall be examined to determine conformance with the requirements of workmanship and dimensions.

4.5.2 Test Conditions - Unless otherwise specified the locator shall be subjected to quality conformance tests under the following conditions:

4.5.2.1 Temperature: 75 ± 5° Fahrenheit.

4.5.2.2 Humidity: To 95 percent, relative, maximum.

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment and facilities, and supply the required personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and of sufficient accuracy and quality to permit satisfactory performance of all required quality conformance tests.

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4.5.4 Test Procedure

NOTES: 1. Test probes or clip leads shall not be used to make electrical connections to any point on the printed circuit board. All test connections to the printed circuit board shall be soldered using the minimum amount of heat necessary to complete or undo the connection.

2. All leads shall be shielded to a maximum and all leads shall be as short as possible.

3. The tests of paragraphs 4.5.4.5 through and including 4.5.4.9 shall be conducted with the control panel mounted in a housing modified to provide access to adjustable capacitors C3 and C12, and test point E-1. A shielded cable shall be connected to the headset terminal and ground, entering the control unit through the unused cable gland on the control panel.

4.5.4.1 Test Site Suitability Tests - During adjustment and testing of the locator, the search head must be placed on rigid nonmetallic supports and shall not be within locating range of any piece of metal. Also, the area must be free of any random electrical noise that will cause a failure of test 4.5.4.1.2. The following additional tests shall be conducted to check the suitability of any proposed adjusting and testing location:

4.5.4.1.1 Position the search head of the locator on its nonmetallic support and place the electronic chassis and headset on a table at least 5 feet away to prevent mutual coupling with the search head. Remove the electronic chassis assembly from its housing and connect the power supply specified in 4.4.13 to the locator using the double male connector (drawing No. 1827516). Connect the a-c voltmeter specified in 4.4.12 to the headset terminals and ground by means of a short (3 feet or less) length of shielded cable. With the locator "OFF-HI-LO" switch on HI, adjust the panel controls for the minimum (i.e., null voltage).

4.5.4.1.2 Slowly sweep the search head over its support. The voltage level shall not vary more than 6 db. Move the electronic chassis assembly on the surface of the table, taking care not to introduce false response indications due to microphonics. The voltage level shall not vary more than 6 db. Remove the test equipment and place the electronic chassis assembly into its housing.

4.5.4.2 Preconditioning Cycle Test - After a 1 hour or longer exposure to an ambient temperature of 75 ± 5°F, place the nonoperating locator (without batteries) into the chamber or cabinet specified in 4.4.5 at a temperature of 160°F (+0, -5° Fahrenheit). Allow a 1 hour ± 5 minutes exposure at 160°F (+0, -5° Fahrenheit). During the second hour ± 5 minute period reduce the temperature linearly to -20°F (+5, -0° Fahrenheit). Allow a 1 hour ± 5 minute exposure at the -20°F temperature. At the end of the -20°F exposure remove the locator from the chamber or cabinet and place it in an ambient temperature of 75 ± 5°F for 1 hour or longer. The locator shall be considered to have passed this test if, after the temperature cycling no cracks in the plastic of the search head or other visual damage to any other component can be observed and if the locator passes the submergence test of 4.5.4.3 and all other performance quality conformance and environmental tests of this specification.

4.5.4.3 Submergence Test - After being exposed to either the preconditioning cycle test specified in 4.5.4.2 or the temperature and humidity test of 4.6.4, the locator (less batteries) shall be completely submerged under natural or simulated sea water, in the pressure tank specified in 4.4.14 at a pressure of 0.9 psi gauge (equivalent to a 2-foot water depth) for 1/2 hour. The pressure then shall be increased to 44.8 psi gauge, corresponding to a 100 foot depth, for a period of 1/2 hour. Reduce the pressure and remove the locator from the tank. The locator shall be considered to have passed this test if, after submergence, it passes the cable leakage resistance test of 4.5.4.4, the performance and quality conformance tests of this specification and if no visual water leaks can be observed inside the locator control unit.

4.5.4.4 Search Head and Cable Leakage Resistance - Unsolder the two transmit-coil leads and their shields from their terminals on the printed circuit board in the control unit. Also unsolder the receive-coil lead and its shield from their terminals. Connect the receive shield and the two shields of the transmit-coil leads to the megohm bridge specified in 4.4.15. The leakage resistance between transmit and receive shields shall be 150,000 megohms or higher. Remove the two shields

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of the transmit-coil leads from the bridge and connect one of the transmit-coil leads to the megohm bridge in their place. The leakage resistance shall be 150,000 megohms or higher. Remove the receive coil shield from the megohm bridge and replace it with the transmit lead shields. The leakage resistance shall be 150,000 megohms or higher. Reconnect the search-head cable leads and shields to their respective terminals on the printed circuit board.

4.5.4.5 Oscillator Frequency - Prior to this test the locator shall be conditioned for 8 hours or more at an ambient temperature of 75 ± 5° Fahrenheit. Place the locator on the nonmetallic supports at the test site. Install the electronic chassis assembly in the modified housing specified in 4.5.4. Connect the power supply specified in 4.4.13 to the locator and adjust the "A" and "B" battery voltages to approximately 1.25 volts and 56 volts respectively and allow a 1/2 hour warm-up period. Connect the frequency meter specified in 4.4.11 between the headset terminal and ground by means of a shielded cable. With the locator "OFF-HI-LO" switch on HI, adjust the R and X panel controls for a minimum (i.e. null voltage) in the headset. Adjust and maintain the "A" and "B" battery voltages to 1.25 ± 0.05 and 56 ± 0.5 volts respectively and re-null the locator. Offset a panel control slightly in order to obtain a higher voltage at the headset terminals sufficient to operate the frequency meter. Adjust the screw-driver control C-3 in incremental steps and record the maximum and minimum frequency observed on the frequency meter. The frequency range shall extend from 2490 cps or less to 2510 cps or more. Readjust C-3 for a frequency of 2500 ± 3 cps.

4.5.4.6 Output Amplifier Tuning - With the electronic chassis assembly installed in the modified housing specified in 4.5.4, connect the a-c voltmeter (4.4.12) between the headset terminal and ground. Adjust the front panel controls for a minimum voltage across the headset. With the locator "OFF-HI-LO" switch set to "HI", place the standardized target pin (4.4.2) over one of the receive coils of the search head. Adjust C-12 for maximum voltage indication. The output shall be 550 ± 100 millivolts.

4.5.4.7 Front Panel Controls - With the electronic chassis assembly installed in a modified housing specified in 4.5.4, adjust the front panel controls for a minimum voltage across the headset and note the positions of the front panel controls. The R front panel control shall be within ±360° from the center of its 5 turn rotational travel, and the X front panel control shall be within ±45° from the center of its rotational travel.

4.5.4.8 Operational Tests - These tests must be conducted in the temperature controlled cabinet or water tank specified in 4.4.4. With the electronic chassis assembly installed in the modified housing specified in 4.5.4, place the chassis and search head on separate rigid nonmetallic supports so that the two units are located at least 5 feet apart. If a water tank is to be used, equip the locator with the special battery compartment cover specified in 4.4.7, and connect a waterproof shielded cable 0.215 ± 0.01 inches O. D. to the headset terminal and ground entering the control unit through the unused cable gland on the control panel. This may be accomplished by removing the plug (drawing No. 1827542) from the extra gland. All other gland parts are used to reseal the cable into the gland. Connect the power supply specified in 4.4.13 to the control unit. Prior to running each test under the conditions specified in table 1, the locator shall be temperature conditioned for 8 hours or more in the chamber or cabinet, or 1/2 hour or more in the water tank with the "OFF-HI-LO" switch turned OFF.

4.5.4.8.1 Test Condition "H" (Table 1) - Temperature condition the nonoperating locator at 75 ± 5°F for at least 8 hours in air or 1/2 hour in water. Turn the "OFF-HI-LO" switch to HI and adjust the "A" and "B" battery voltages to 1.25 ± 0.05 volts and 56 ± 0.5 volts, respectively. Allow the operating locator to warm up electronically for 1/2 hour or more.

4.5.4.8.1.1 Connect the a-c voltmeter specified in 4.4.12 between the headset terminal and ground via the shielded cable. Adjust the panel controls for the minimum rms voltage as indicated on the voltmeter, and record the voltage reading. The minimum rms voltage shall not exceed 40 millivolts

4.5.4.8.1.2 Place the standardized target pin (4.4.2) over the right receive coil and record the headset voltage reading. The rms voltage shall be between 450 and 650 millivolts.

4.5.4.8.1.3 Move the standardized target pin to the left receive coil and record the voltmeter reading. Thr rms voltage shall be within ±1 db of that measured in 4.5.4.8.1.2 when the standardized target pin (4.4.2) was on the right receive coil.

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4.5.4.8.1.4 Remove the standardized target pin (4.4.2) from the receive coils and slowly bring any large saturating target close to either end of the search head. The rms voltmeter reading shall increase by a total of 10 db or more over that recorded in either 4.5.4.8.1.2 and 4.5.4.8.1.3. As the target is moved toward the search head, the voltage shall increase to a maximum at the search head or just prior to reaching the search head. If the maximum voltage occurs at some position of the target other than at the search head, the voltage with the target positioned on the search head shall not be more than 0.5 db below the maximum level observed. Remove the saturating target.

4.5.4.8.1.5 Turn the "OFF-HI-LO" switch to "LO" and adjust the front panel controls for a minimum headset voltage. Place the standardized target pin (4.4.2) over the right receive coil. The rms voltage level shall be 30 to 80 millivolts. Remove the standardized target pin (4.4.2).

4.5.4.8.1.6 Turn the "OFF-HI-LO" switch to "HI" and adjust the front panel controls for a minimum headset voltage. Place the phase test block specified in 4.4.3 over the right receive coil and record the headset voltmeter reading. The rms voltage shall be 7 db or more below the voltage measured in 4.5.4.8.1.2.

4.5.4.8.1.7 Move the phase test block (4.4.3) to the left receive coil and record the headset voltmeter reading. The rms voltage shall be 7 db or more below the voltage measured in 4.5.4.8.1.3 and within 2 db of the voltage measured in 4.5.4.8.1.6. Remove the phase test block.

4.5.4.8.1.8 Adjust the front panel controls for a minimum headset voltage. Check the stability of the locator over a 15 minute period by monitoring the headset voltage. The rms voltage shall not exceed 80 millivolts over the 15 minute period.

4.5.4.8.2 Test Conditions "G" and "I" (Table 1) - Repeat the tests specified in 4.5.4.8.1.1 through 4.5.4.8.1.3, 4.5.4.8.1.6, and 4.5.4.8.1.7 (deleting the ±2 db requirement of 4.5.4.8.1.7). The locator shall meet the requirements for the respective test conditions listed in table 1. Allow 1/2 hour electrical warmup at specified input voltages before testing.

4.5.4.9 Electronic Circuit Test - With the electronic chassis assembly installed in the modified housing specified in 4.5.4, connect the power supply specified in 4.4.13 to the locator electronic chassis assembly. Turn the "OFF-HI-LO" switch to OFF, temperature condition the locator at 75 ± 5°F for 8 hours or more, and then conduct the tests of 4.5.4.9.1 at that temperature.

4.5.4.9.1 Second Harmonic Level - Connect the a-c voltmeter specified in 4.4.12 to the headset terminal and ground. Turn the "OFF-HI-LO" switch to HI and adjust and maintain the "A" and "B" battery voltages at 1.25 ± 0.05 volts and 56 ± 0.5 volts, respectively. Allow the unit to warm up for 30 minutes or more. Adjust the locator front panel controls for the minimum voltage (i.e. null voltage) measured at the headset terminals. Connect the a-c VTVM (4.4.8) to test point E-1. The voltage measured within the VTVM (4.4.8) between terminal E-1 and ground shall not exceed 200 millivolts rms. Failure to meet this requirement may be due to a faulty variometer plug-in unit Z-10. The same test is required in an operating locator for each spare variometer plug-in unit (Z-10) that is produced.

4.5.4.10 Nonmagnetic Tests for a Nonoperating Locator -

NOTE: Reference paragraph 2.5, of NAVORD OD 10511 for evaluation of magnetic signals produced by the following tests.

4.5.4.10.1 Headset -

4.5.4.10.1.1 Headband - The headband shall be tested using the procedure of Test Method I, Orientations A, B, and C of NAVORD OD 10511. The test distance shall be 1-3/4 ± 1/8 inches. For orientations A and B, move the headband until all parts of it are centered over the magnetometer, with the convex surface towards the magnetometer. For orientation C, the test face is the surface tangent to the outside center of the headband. The maximum permissible signal obtained for any of the three test cycles shall not exceed 0.05 milligauss.

4.5.4.10.1.2 Headphones - Each headphone shall be separated from the headband and tested individually using the procedure of Test Method I of NAVORD OD 10511, Orientation A only. Before

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beginning the test, the second headphone shall be separated from the headphone to be tested by the length of the connecting cable, about 16 inches, by moving it away from the test position in a horizontal direction. The second headphone should be attached to a support and not allowed to move during the test. The test distance shall be 6 ± 1/8 inches. Six test faces shall be used as follows: Imagine the headphone circumscribed by a rectangular parellelepiped, disregarding the outer 1/2 inch of the rubber disc. Each face of the parallelepiped is then a test face, one face being the outer flat surface of the plastic housing. The maximum permissible signal obtained for any of the six test cycles for each headphone shall be 0.4 milligauss.

4.5.4.10.2 The Search Head Static Nonmagnetic Test - Prior to conducting this test, the nylon nut holding the short handle to the search head shall be removed and the short handle together with the extension handle (collapsed condition) shall be pushed up the cable to the control unit. During the magnetic test all other components should be kept as far away from the magnetometer as the search head cable permits.

4.5.4.10.2.1 The nonmagnetic test shall be conducted in each of the following three positions in accordance with the procedure specified in NAVORD OD 10511, Test Method 1, Orientations A, B, and C. Position (a) is with the major ellipse axis at the bottom of the search head maintained at a distance of 3 ± 0.5 inches from the center of the magnetometer. Position (b) is with the major ellipse axis of the top of the search head maintained at a distance of 3 ± 0.5 inches from the center of the magnetometer. For each position (a) and (b), the axis should be swept past the magnetometer in a 5 to 10 second time interval. Position (c) is with the magnetometer in the plane of the search head. The distance from the edge of the search head to the center of the magnetometer shall be maintained at 3 ± 0.5 inches, while the circumferential edge of the search head is moved past the magnetometer. The maximum permissible signal obtained on any test shall not exceed 0.10 milligauss peak-to-peak.

4.5.4.10.3 The Handle Nonmagnetic Test - Prior to conducting the test the short handle shall be mechanically disconnected from the search head (as in 4.5.4.10.2). The short handle coupled to the extension handle (collapsed condition) shall be positioned along the cable midway between the search head and the control unit.

4.5.4.10.3.1 The short handle coupled to the extension handle (collapsed condition) shall be subjected to the test procedure specified in NAVORD OD 10511 Method 1. The short handle coupled to the extension handle (collapsed condition) shall be swept longitudinally, at a distance of 3 ± 0.5 inches from the center of the magnetometer in a time interval of 5 to 10 seconds. Rotate the handle 90° around its axis and repeat the above test. The maximum permissible signal obtained in each test shall not exceed 0.10 milligauss peak-to-peak.

4.5.4.10.4 Control Unit Nonmagnetic Test

4.5.4.10.4.1 Magnetic Idealization Procedure - Prior to conducting this test the control unit shall be subjected to the magnetic idealization test as specified in NAVORD OD 10511. For this test the control unit shall be positioned so that the 5 gauss magnetic field passes through the control unit along the axis between the belt buckles on one face and the nameplate on the opposite face.

4.5.4.10.4.2 Nonmagnetic Test - Following the magnetic idealization test above, the control unit shall be subjected to the nonmagnetic test as specified in NAVORD OD 10511, Test Method I. The test shall be conducted for each face (six faces total) of the control unit at a distance of 6 ± 0.5 inches from the center of the magnetometer. The distance to the control panel surface shall be used for the measurements of 6 ± 0.5 inches on the one face rather than the outer surfaces of the knobs. The maximum permissible signal obtained on any face shall not exceed 4.0 milligauss peak-to-peak.

4.5.4.10.4.3 Demagnetization Procedure - Repeat the magnetic idealization test 4.5.4.10.4.1 with 5 gauss background magnetic field switched off.

4.5.4.10.4.4 Nonmagnetic Tests - Repeat the nonmagnetic test of 4.5.4.10.4.2. For this test the maximum permissible signal obtained shall not exceed 1.0 milligauss peak-to-peak.

4.5.4.10.5 Belt Nonmagnetic Test - The belt nonmagnetic test shall be conducted in accordance with NAVORD OD 10511 Test Method I. The belt shall be stretched taut and either of the two flat sides swept along its longitudinal axis at a distance of 3 ± 0.5 inches from the center of the

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magnetometer in a time interval of 10 to 15 seconds. The test shall be repeated using either edge of the belt and sweeping longitudinally, at a distance of 3 ± 0.5 inches from the center of the magnetometer. The maximum permissible signal obtained on any of the above tests shall not exceed 0.10 milligauss peak-to-peak.

4.5.4.11 Spare Plug-In Units Test - Install all spare plug-in units in the locator and test to insure that the locator shall meet all the performance requirements of this specification

4.5.4.12 Search Head Tests - The search head shall be tested in accordance with drawing No. 1698359 and 4.5.4.10.2.

4.5.4.13 Headset - The headset shall be tested in accordance with drawing No. 2293749 and 4.5.4.10.1.

4.6 Preproduction and Periodic Production Tests

4.6.1 Vibration, Transportation; Storage, and Handling - For this test the locator, in its transit case, shall be mounted rigidly to a vibration table. The locator shall be mounted alternately in three mutually perpendicular orientations such that it experiences vibration for one-third of the duration of each frequency increment in each orientation. The locator shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in increments of 100 cycles per minute. The duration of vibration of each increment shall be 1 hour. The amplitude (zero to peak) of the vibration shall be in accordance with the following schedule:

FREQUENCY (Cycles per minute)	AMPLITUDE (Inches zero-to-peak)
700 to 1500	0.030 ± 0.002
1600 to 2000	0.020 ± 0.002
2100 to 2500	0.013 ± 0.001
2600 to 3000	0.009 ± 0.001

4.6.2 Repeated Impact; Transportation, Storage, and Handling - For this test the locator, in its transit case, shall be placed on a vibration platform equipped with barriers so as to allow free movement of the locator relative to the platform but prevent it from falling off the platform. The motion of the platform shall be such that any point on the platform moves in a circular path 1 inch in diameter with the plane of the circle perpendicular to the platform. The frequency shall be within the range of 240 to 300 cycles per minute. The vibration shall be for a period of 1/2 hour on each face of the container.

4.6.3 Shock; Transportation, Storage, and Handling - The locator, in its transit case, shall be dropped cornerwise from a height of 30 inches onto a solid steel or concrete surface in such manner that each corner of the case absorbs the full force of the fall. This test shall be applied to each of the eight corners of the case. (The height of 30 inches refers to the distance from the steel or concrete surface to the nearest edge of the locator container when suspended prior to fall.) The fall shall be freefall, in that no ropes or other suspending media are attached to the container during the fall.

4.6.4 Temperature and Humidity Test - The locator, (with the batteries removed), shall be subjected to a temperature and humidity 28-day cycle in accordance with MIL-STD-331, , as modified to restrict the low temperature to -20°F instead of -65° Fahrenheit.

4.6.5 Submergence and Leakage Resistance Tests - The locator (with the batteries removed) shall be subjected in the following order to the temperature cycling test of 4.6.4, the sea water submergence test of 4.5.4.3, and to the search head and cable leakage resistance test of 4.5.4.4, after which it shall meet all the performance and other requirements of this specification.

4.6.6 Altitude - For this test the locator in its transit case shall be placed in a test chamber and the pressure reduced to 1.7 psia ± 0.2 psia, corresponding approximately to 50,000 feet

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altitude, for a minimum period of 15 minutes. Following this test the locator shall be operable and shall meet the performance requirements of this specification.

4.6.7 Salt Spray - The locator, removed from its transit case, plus the open transit case and cover, shall be subjected to the salt spray corrosion test of MIL-STD-202, Method 101A. The salt solution concentration shall be 5 percent. Time of test shall be 96 hours (Test Condition A). Following the test the locator and transit case shall be examined for corrosion which would adversely affect operation or life.

4.6.8 Cable Cold Bend Submergence Test - For this test each 10-ft. sample of cable submitted as specified in 4.1.1 and 4.1.3 shall be cut into two 5-ft. specimens. One specimen shall be subjected to the cold bend test. The specimen to be tested first shall be wound around a 3 inch diameter wooden mandrel. The mounted specimen on the mandrel shall be placed in a refrigerator and conditioned at -20 ± 5°F for 48 hours or more. The specimen then shall be unwound from the mandrel while in the refrigerator, at a uniform rate of approximately one revolution per second. The test cable shall then be subjected to the submergence test of 4.5.4.3 (omitting the temperature cycle). Search head cables shall be subjected to the leakage resistance tests of 4.5.4.4. Headset cables shall be tested for leakage resistance using megohmmeter of 4.4.15 between the lead and shield. The minimum acceptable level is 150,000 megohms. The specimen shall then be dissected and a visual inspection made of all its component parts. When defects or derangements are discovered in any component, examination should be made of the specimen taken from the remainder of the sample in the as-received condition to determine whether the defect existed prior to the cold bend test. Any failures of the submergence or leakage resistance tests or any derangements or damage resulting from the cold bend test, such as torn tapes or broken or cracked components shall be cause for rejection.

4.6.9 Control Unit Temperature Stability - For these tests the locator shall be equipped with the special battery compartment cover drawing No. 2007343 specified in 4.4.7. Connect the locator to the power supply specified in 4.4.13. Connect the a-c voltmeter specified in 4.4.12 to the headset terminal and ground by means of an external shielded cable passed through the spare cable waterproofing gland on the front panel. Turn the "OFF-HI-LO" switch to HI and adjust the "A" and "B" battery voltages to 1.25 ± 0.05 volts and 56 ± 0.5 volts, respectively.

4.6.9.1 Adjust the water temperature in one of the test tanks specified in 4.4.6 to 120 ± 2°F and in the other tank to 110 ± 2° Fahrenheit. Submerge the locator control unit only in the 120°F tank for 30 minutes or more. The search head shall be maintained at an ambient temperature of 75 ± 5° Fahrenheit. Adjust the locator front panel controls for a minimum voltage (i.e. null voltage) and record the level. Quickly (less than 2 seconds) remove the locator control unit from the 120°F tank and plunge it into the 110°F tank. During the first 5 ± 0.5 minute period at 110°F, note and record the maximum voltage change, which shall not exceed 8 db. Remove the locator control unit from the tank.

4.6.9.2 Repeat the test procedure of 4.6.9.1 using tank temperatures of 70 ± 2°F and 60 ± 2° Fahrenheit. The maximum voltage change shall not exceed 8 db.

4.6.9.3 Repeat the test procedure of 4.6.9.1 using tank temperatures of 42 ± 2°F and 32 ± 2° Fahrenheit. The maximum voltage change shall not exceed 8 db. Remove all test equipment and reseal the spare cable waterproofed gland using plug Drawing No. 1827542.

4.6.10 Operational Tests - Perform the tests of 4.5.4.8 through 4.5.4.8.1.8 for each of the test conditions of table I. The requirements of 3.3.5 through 3.3.5.8 apply.

4.6.11 Electronic Circuit Tests - For these tests the electronic chassis assembly shall be removed from its housing. Connect the power supply specified in 4.4.13 to the locator electronic chassis assembly using the double male connector drawing No. 1827516. Turn the "OFF-HI-LO" switch to OFF, temperature condition the locator at 75 ± 5°F for 8 hours or more, and then conduct the tests of 4.6.11.1 through 4.6.11.3 at that temperature.

4.6.11.1 Frequency Response Output Amplifier and Transformer - Remove plug-in units Z2 and Z-6 from their sockets. Connect the output of the microvolter specified in 4.4.9 to test point E-1 by means of a short length of well shielded cable. Connect the audio signal generator specified

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in 4.4.10 to the input of the microvolter. Connect the a-c voltmeter specified in 4.4.12 between the headset terminal and ground. Connect the frequency meter specified in 4.4.11 to the output of the signal generator. Turn the locator "OFF-HI-LO" switch to HI. Adjust and maintain the "A" and "B" battery voltages to 1.25 ± 0.05 volts and 56 ± 0.5 volts respectively, and warm up the locator electrically for 30 minutes or more. Maintain the output of the microvolter at 5 mv and slowly vary the frequency of the signal generator from 1000 to 4000 cps. Raise the microvolter output to 50 mv and slowly vary the frequency of the signal generator from 4,000 to 10,000 cps. Record the frequency and headset voltage for every maximum and minimum voltage point observed over the total frequency range. The response at 5000 cps shall be 60 db or more below the response at 2500 cps when based on the 5 mv input level. Turn the "OFF-HI-LO" switch to OFF. Remove the signal generator, the a-c voltmeter, the microvolter, and the frequency meter.

4.6.11.2 Crosstalk Measurements - Connect the a-c voltmeter specified in 4.4.12 between the headset terminal and ground. Set the locator "OFF-HI-LO" switch to HI and the "A" and "B" battery voltages at 1.25 ± 0.05 volts and 56 ± 0.5 volts respectively. Short test point E-1 ground. The voltmeter reading shall not exceed 10 mv. Remove the short from test point E-1 and short pin 8 of Z-6 to ground. The voltmeter reading shall not exceed 500 mv. Remove unit Z-7 from its socket. The voltmeter reading shall not exceed 500 mv. Remove Z-6 from its socket. The voltmeter reading shall not exceed 300 mv. Replace Z-7 in its socket. The voltmeter reading shall not exceed 300 mv. Replace Z-6 in its socket and remove the short to ground on pin 8 of Z-6. Short pin 1 of Z-5 to ground. The voltmeter reading shall not exceed 1.0 volt. Remove all of the test equipment and all shorts and replace the electronic chassis into its housing. See soldering note in 4.5.4.

4.6.12 Additional Preproduction Tests - In addition to the tests of 4.6.1 through 4.6.11.2, the preproduction samples shall also be subjected to all quality conformance tests of 4.5 through 4.5.4.13 and shall be inspected for conformance to all other requirements of Section 3 (Requirements).

4.7 Acceptance Criteria - Sample units which fail to comply with any of the test or inspection requirements shall be classified as defective. If the number of defective units exceeds the number of defectives permitted by the associated acceptance criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Not applicable.

5.2 Packing

5.2.1 Level A - Not applicable.

5.2.1.1 Exterior Container - Each locator assembly shall be packed in an aluminum chest with specially designed cushions conforming to the requirements of Bureau of Ordnance LD No. 496307.

5.2.1.2 For arrangement of locator assemblies and details, see Bureau of Ordnance Drawing No. 1875134.

5.2.2 Level B - Not applicable.

5.2.3 Level C - Not applicable.

5.3 Marking

5.3.1 Special Markings - Not applicable.

5.3.2 Normal Markings - In addition to any special marking required by the contract or order, all marking shall be in accordance with MIL-STD-129, including serial numbers.

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5.3.2.1 For identification and destination marking, a fiberboard pad, cut to fit within the rim of the cover, shall be positioned on the outside of the aluminum chest, MIL-C-16775, Size 1. Fiberboard shall meet the material requirements of Class 2, W6c or W6s compliance symbol conforming to the requirements of Federal Specification PPP-B-636.

5.3.2.2 Two flat steel straps shall be employed to prevent removal of the fiberboard pad insert and also provide additional closure protection. Strapping shall be in accordance with Federal Specification QQ-S-781.

6. NOTES

6.1 Intended Use - The locator is intended to locate buried ordnance on land or submerged in water to a depth of 100 feet.

6.2 Ordering Data - Procurement documents shall specify Level A packaging and packing requirements for all shipments.

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